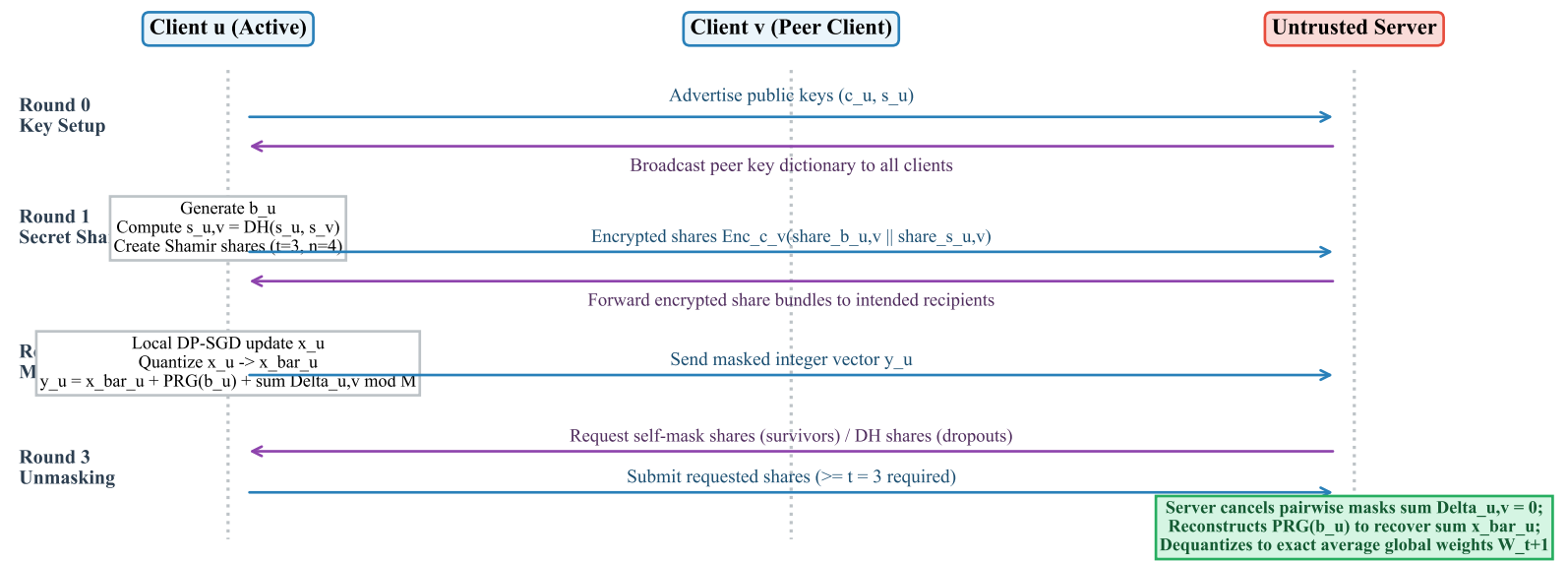
Client u (Active)
Client v (Peer Client)
Untrusted Server
Round 0
Key Setup
Advertise public keys (c_u, s_u)
Broadcast peer key dictionary to all clients
Round 1
Secret Sha
Generate b_u
Compute s_u,v = DH(s_u, s_v)
Create Shamir shares (t=3, n=4)
Encrypted shares Enc_c_v(share_b_u,v || share_s_u,v)
Forward encrypted share bundles to intended recipients
R
M
Local DP-SGD update x_u
Quantize x_u -> x_bar_u
y_u = x_bar_u + PRG(b_u) + sum Delta_u,v mod M
Send masked integer vector y_u
Round 3
Unmasking
Request self-mask shares (survivors) / DH shares (dropouts)
Submit requested shares (>= t = 3 required)
Server cancels pairwise masks sum Delta_u,v = 0;
Reconstructs PRG(b_u) to recover sum x_bar_u;
Dequantizes to exact average global weights W_t+1